Technology Stack & Architecture Matrix

Date	19 September 2026
Team ID	
Project Name	FreelanceHub — MERN Stack Freelancing Platform
Maximum Marks	4 Marks

Table 1: Components & Technologies

S.No	Component	Description	Technology
1	**User Interface**	Interactive single-page application handling client-side routing, role-based dashboards for clients and freelancers, project filtering, and toast notifications.	React.js, Vite, Bootstrap, React Router DOM, React Toastify
2	**Application Logic — Frontend API Layer**	A shared client service layer handling all HTTP communication with the backend, with the JWT automatically attached to each request.	Axios
3	**Application Logic — Global State**	Stores the logged-in user, role, and token so every component knows who is logged in and what they can access.	React Context API (Auth Context), localStorage
4	**Application Logic — Backend Services**	Handles routing, request validation, and business logic for authentication, projects, and applications.	Node.js, Express.js
5	**Application Logic — Security**	Authenticates users, hashes passwords, and issues/verifies signed session tokens; restricts routes by role (Client / Freelancer / Admin).	JSON Web Token (JWT), bcryptjs
6	**Real-time Communication**	Opens a persistent two-way connection so approved clients and freelancers can exchange chat messages instantly.	Socket.io (server), socket.io-client (frontend)
7	**Database**	Document-oriented storage for users, freelancer profiles, projects, applications, and chats.	MongoDB (NoSQL Document Store)
8	**Object Data Mapping**	Schema definition, validation, and query interface between the Express backend and MongoDB.	Mongoose ODM, MongoDB Atlas (cloud-hosted cluster)
9	**Infrastructure / Deployment**	Runtime environment for both frontend and backend during development, with source code managed on GitHub.	Node.js runtime (v16+); currently run locally via npm run dev — not yet deployed to a public host; Git & GitHub for version control

Table 2: Application Characteristics

S.No	Characteristic	Description	Technology Implementation
1	**Open-Source Frameworks**	Every layer of the stack — frontend, backend, database, and real-time messaging — is built on freely available, widely-adopted open-source technology.	MERN Stack (MongoDB, Express, React, Node.js) + Socket.io
2	**Security Implementation**	Passwords are never stored in plain text; sessions are stateless and verified on every protected request, and each role can only reach its own routes.	bcrypt password hashing, server-signed JWTs, protect/authorize role-based middleware, ProtectedRoute on the frontend

S.No	Characteristic	Description	Technology Implementation
3	**Scalable Architecture**	A clean three-layer separation (routes/middleware, controllers, models) means new features such as payments or reviews can be added without touching existing code paths.	MVC pattern via Express + Mongoose
4	**Data Integrity**	Every application is linked to a valid project and freelancer, and statuses are limited to fixed values, keeping project and application records consistent.	Mongoose schema validation, ObjectId references between collections, enum status values (Pending / Approved / Rejected)
5	**Performance**	Client-side routing avoids full page reloads, and chat messages are pushed instantly instead of repeatedly asking the server for updates.	React Router SPA navigation, Socket.io event-based messaging, filtered MongoDB queries